

React Fun with useState & Event Handling

Interactive Frontend Development • Beginner's Study Guide & Notes

1. Understanding State & Events

In standard websites, content is static—like a printed dice that can't roll. In modern web applications (like React apps), we want interactive **LIVE** components—like a clickable live dice or an interactive scoreboard!

What is "State"? (Your Component's Personal Notebook)

State is your React component's live memory box or digital piggy bank. It stores values that change over time as users interact with the app (e.g., cricket score, fan cheers, money in a piggy bank).

What are "Events"? (Calling Bell or Kabaddi Out Call)

Events are actions triggered by the user—like clicking a button, hovering, or pressing a key. In React, events signal your code to perform an action and update the state.

2. Syntax & Project: Live IPL Fan Cheer Counter

Let's build a Live IPL Fan Cheer Counter! Every tap on the button adds a cheer to our live state memory.

```
// Step 1: Don't forget to import useState!
import React, { useState } from 'react';

function CheerCounter() {
  // Syntax: const [stateValue, updaterFunction] = useState(initialValue);
  const [cheers, setCheers] = useState(0);

  return (
    <div>
      <h1>Live Cheers: {cheers}</h1>

      // Button triggers event to update state (like a metro card tap!)
      <button onClick={() => setCheers(cheers + 1)}>
        +1 Cheer 🎉
      </button>

      <button onClick={() => setCheers(0)}>
        Reset 🔄
      </button>
    </div>
  );
}
```

```
    </div>  
  );  
}
```

3. Bug Buster: Avoid Common Oopsies! 🐛

⚠️ 1. Capital 'C' in onClick

React uses **camelCase** for event handlers! Writing `onClick` (lowercase) will not work. Always use `onClick`.

⚠️ 2. Parentheses for State Setter

Always call your updater function with parentheses e.g. `setCheers(cheers + 1)`. Never assign directly like `cheers = cheers + 1`!

💡 3. Always Import Hook

Make sure you include `import { useState } from 'react'`; at the very top of your component file.

Student Practice Sheet

Test Your Knowledge: React useState & Event Handling

Question 1

Multiple Choice

What happens in React if you spell the click event listener as `onclick` (with a lowercase 'c') instead of `onClick`?

- A) It works perfectly fine without any issues.
- B) Nothing happens / It is ignored because React requires camelCase event names.
- C) It throws a critical syntax error that crashes the browser instantly.
- D) Your computer explodes into fireworks!

Question 2

Conceptual

In our lesson analogy, `useState` was compared to a "Personal Notebook" or "Magic Piggy Bank". In your own words, explain why state is necessary in a dynamic React component.

Question 3

Code Analysis

Look at the following line of code: `const [runs, setRuns] = useState(0);`. Identify what each of the three components (`runs`, `setRuns`, and `0`) represents.

Question 4

Bug Fixing

A student wrote the following React button code to increment a score, but it isn't working properly:

```
<button onclick={setRuns = runs + 1}>Add Run</button>
```

Identify two distinct bugs in the student's code snippet above.

Question 5

Mini-Project Challenge

Write a small React component named `DiwaliDiya` that maintains a boolean state `isOn` (starting at `false`). Include a button that toggles the state between ON and OFF, and display text showing "🔥 Diya is Glowing!" when ON, and "🌕 Diya is Off" when OFF.